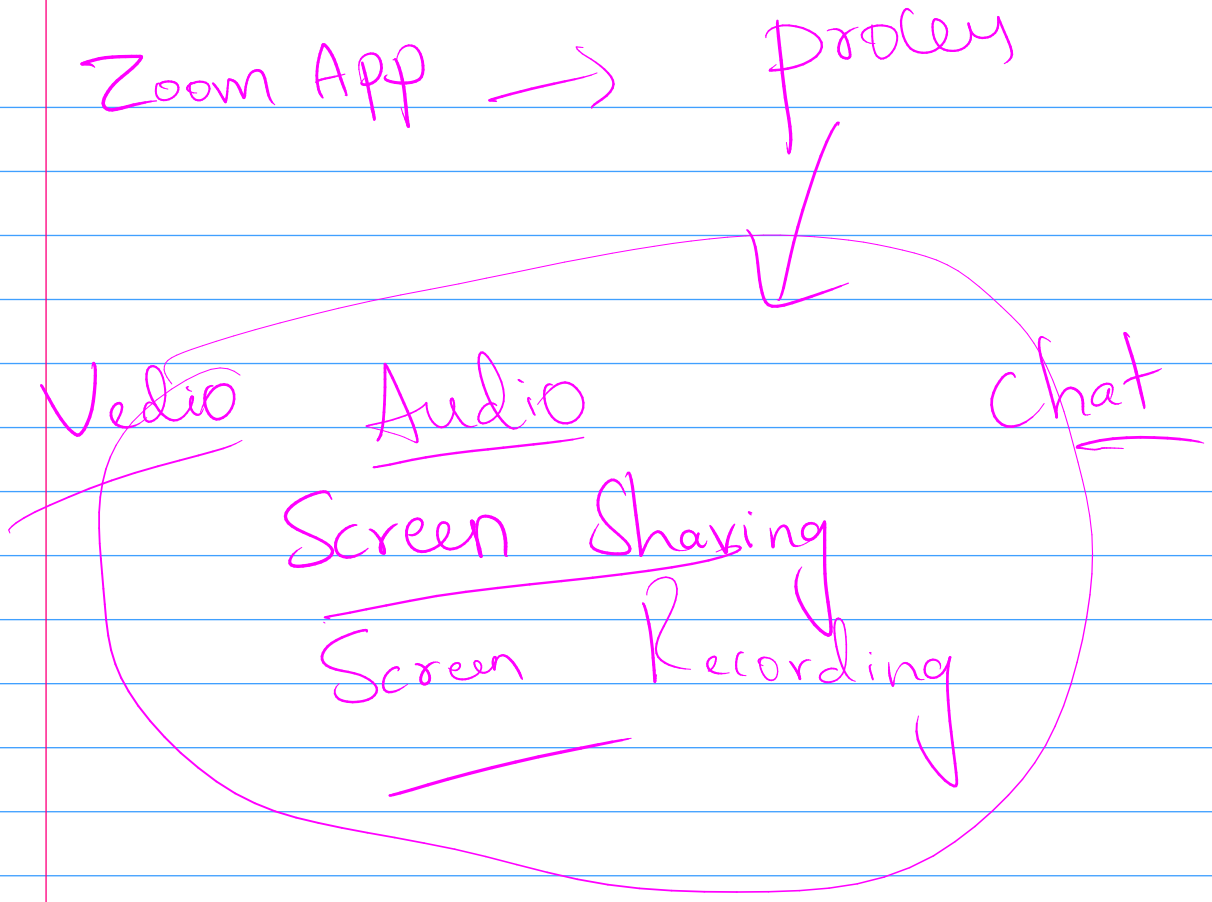


Teams

Whatsapp

Desktop/Laptop

Hex file → Flash magic → Hw Target board

Thread → TID → further operation

TCB → Thread Control block

Stack

pc

PCB



P1 → T1 & T2

TCB of T1

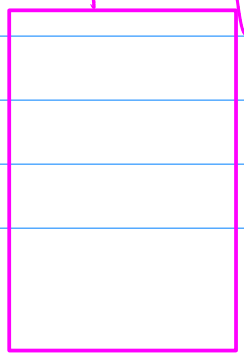
pc

SP

cpu reg

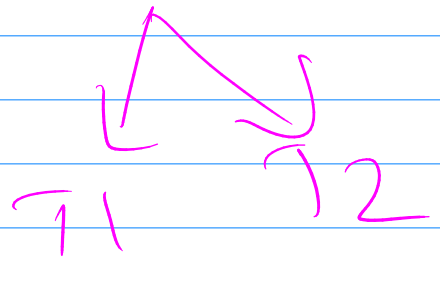
ptr → PCB

TCB of T2



→ TCB → It is a data structure containing the infⁿ about the specific thread and a ptr to its respective PCB.

→ process is Scheduled



↓ Scheduled

PC → T1 entry point : default

CPU reg

SP

Current thread states is stored into its TCB & newly Scheduled

thread states is loaded. and it is called thread Context Switching.

→ Thread Context Switching time is less Compare to process Context Switching ie thread is called as light weighted process.

Process Vs Thread

process	Thread
Program undy execution	Basic unit of execution.
→ process Can be of Single threaded or multi threaded.	→ Light weighted process.
→ fork / Vfork System Call.	Can be created → Library fns like pthread_create

→ process will Seperate address space	→ Threads of a process consists of Seperate address space Stack section
→ Terminating a process terminates all the threads of a process	→ terminating a thread will not effect on other thread.
→ exit(0) exit(0)	→ pthread_exit(NULL)

→ Thread related functions:-

pthread_create

pthread_join

pthread_exit

gettid()

pthread_create :-

ptr → add var

```
main()
{
    int a, b;
    ≡
    f1(a, b);        f1(&a, &b)
    ≡
}
void f1(int a, int b)        int *p, int*
{
    ≡
}
```

```
main()
{   int a,b;
    p = &a
    q = &b
    f1(p, q)
}
```

```
void f1(int *a, int *b)
{
    ≡
}
```

```
void Sum(int a, int b)
{
  ==
  ==
}

void Sub(int a, int b)
{
  ==
  ==
}

void f1(int x,
        void (*fp)(int a, int b)
{
  ==
}

main( )
{
  f1(3, Sub)
  f1(4, Add)
```

void (*fp) (int, int);

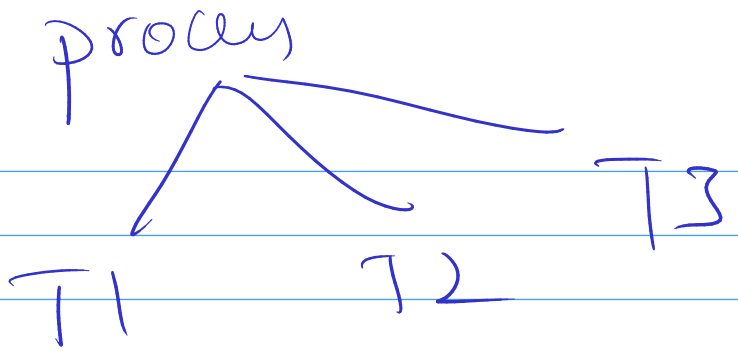
↓

fp (function ptr) is going to point a fn having arg as int & int then return type as void.

Call back code :-

↳ It is a code (function) passed as an argument to another code (fn) and called some later.

→ Call back code is collected with fn ptr.



NULL → default attributes

Stack reserved is default size (2MB)

Thread as joinable typ

detachable type

joinable type → gets terminated → exit states

another thread → pthread_join

T1 T2 T3

pthread_join (tid)

→ Thread entry pt (call back code)
Collected fn pointer.

4th arg :-
argument to be passed when a thread is scheduled (execution starts).